

Student Performance Analysis

Using NumPy, Pandas, and Matplotlib

Dataset: Students Performance (Kaggle)

Total Students: 1000

Generated with Python

Part A - Loading and Exploring Data

1. First 5 Rows

gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
female	group B	bachelor's degree	standard	none	72	72	74
female	group C	some college	standard	completed	69	90	88
female	group B	master's degree	standard	none	90	95	93
male	group A	associate's degree	free/reduced	none	47	57	44
male	group C	some college	standard	none	76	78	75

2. Last 5 Rows

gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
female	group E	master's degree	standard	completed	88	99	95
male	group C	high school	free/reduced	none	62	55	55
female	group C	high school	free/reduced	completed	59	71	65
female	group D	some college	standard	completed	68	78	77
female	group D	some college	free/reduced	none	77	86	86

3. Dataset Dimensions

Number of rows: 1000

Number of columns: 9

4. Column Names

- gender
- race/ethnicity
- parental level of education
- lunch
- test preparation course
- math score
- reading score
- writing score
- Average Score

5. Data Types

Column	Data Type
gender	str
race/ethnicity	str
parental level of education	str

Column	Data Type
lunch	str
test preparation course	str
math score	int64
reading score	int64
writing score	int64
Average Score	float64

Part B - Data Cleaning

1. Missing Values Check

Total missing values: 0

Column	Missing Count
gender	0
race/ethnicity	0
parental level of education	0
lunch	0
test preparation course	0
math score	0
reading score	0
writing score	0
Average Score	0

2. Duplicate Rows

Rows before removing duplicates: 1000

Rows after removing duplicates: 1000

3. Verification

Missing values: 0

Duplicate rows: 0

Dataset is clean and ready for analysis.

Part C - NumPy Analysis

NumPy Arrays Created

math score array shape: (1000,)
reading score array shape: (1000,)
writing score array shape: (1000,)

Statistical Summary

Statistic	Math	Reading	Writing
Mean	66.09	69.17	68.05
Median	66.00	70.00	69.00
Maximum	100	100	100
Minimum	0	17	10
Std Deviation	15.16	14.59	15.19

Average Score Column

A new column "Average Score" was created using np.mean() across all three subject scores for each student.

Part D - Data Analysis with Pandas

1. Student with Highest Average Score

Index: 458

Gender: female

Race/Ethnicity: group E

Parental Education: bachelor's degree

Math: 100, Reading: 100, Writing: 100

Average Score: 100.00

2. Subject with Highest Average Marks

Math: 66.09

Reading: 69.17

Writing: 68.05

Highest: Reading (69.17)

3. Students Scoring Above 90 in Mathematics

Number of students: 50

4. Average Score by Gender

Gender	Math	Reading	Writing	Average
female	63.63	72.61	72.47	69.57
male	68.73	65.47	63.31	65.84

5. Average Score by Parental Level of Education

Parental Education	Math	Reading	Writing	Average
associate's degree	67.88	70.93	69.90	69.57
bachelor's degree	69.39	73.00	73.38	71.92
high school	62.14	64.70	62.45	63.10
master's degree	69.75	75.37	75.68	73.60
some college	67.13	69.46	68.84	68.48
some high school	63.50	66.94	64.89	65.11

6. Lunch Type Comparison

Lunch Type	Math	Reading	Writing	Average
free/reduced	58.92	64.65	63.02	62.20
standard	70.03	71.65	70.82	70.84

Higher average marks: "standard" lunch type (70.84)

7. Test Preparation Course Comparison

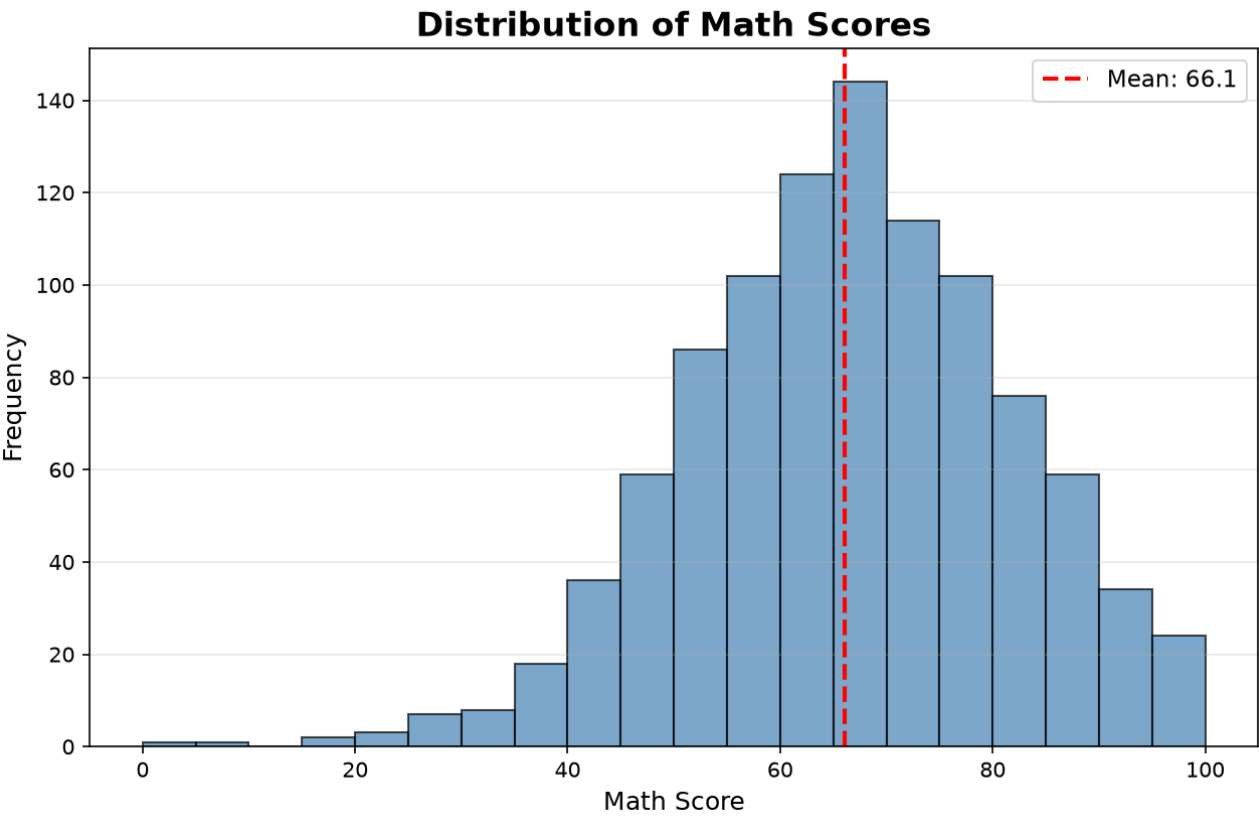
Test Prep	Math	Reading	Writing	Average
-----------	------	---------	---------	---------

Test Prep	Math	Reading	Writing	Average
completed	69.70	73.89	74.42	72.67
none	64.08	66.53	64.50	65.04

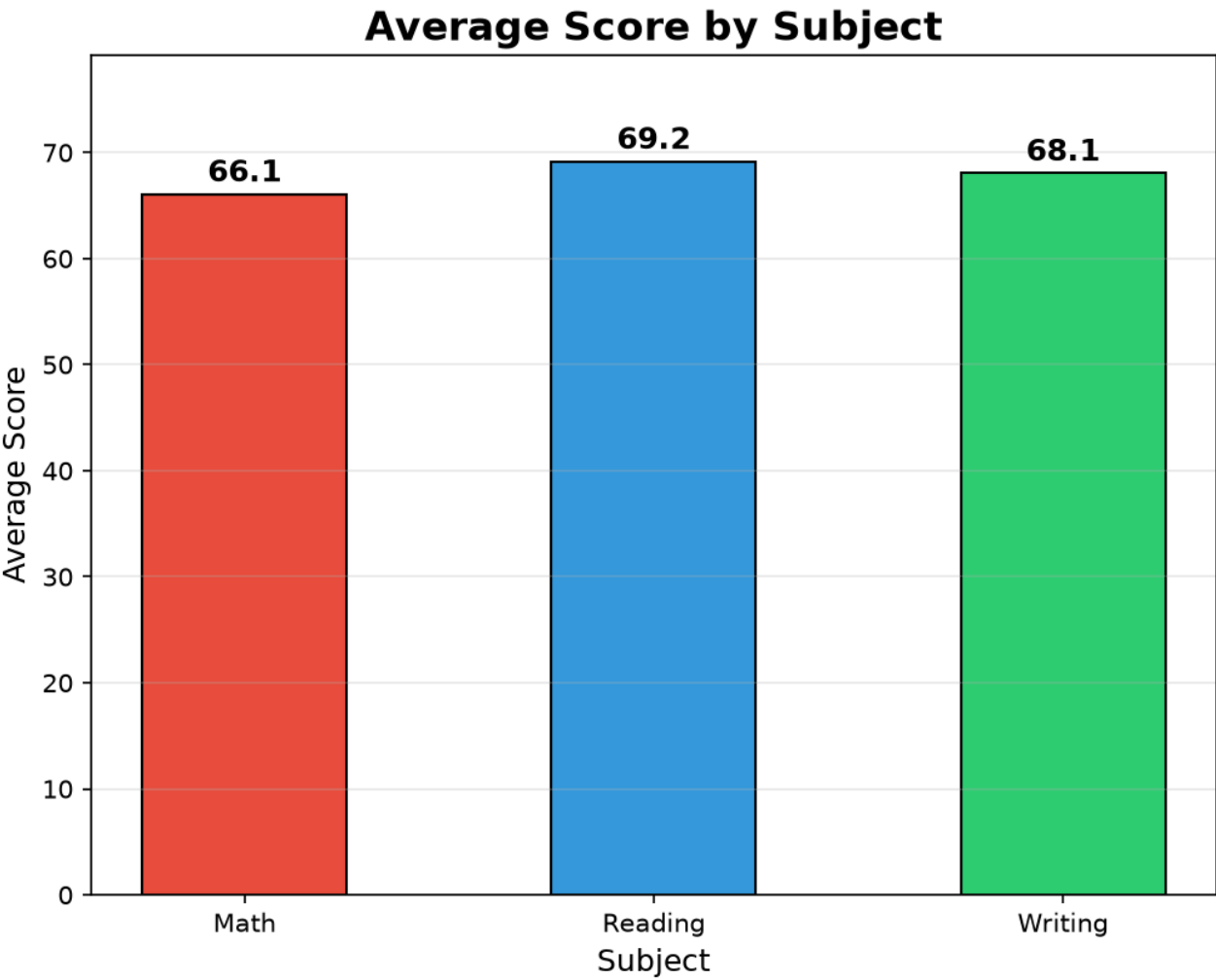
Completed test prep scored higher: 72.67 vs 65.04

Part E - Visualization

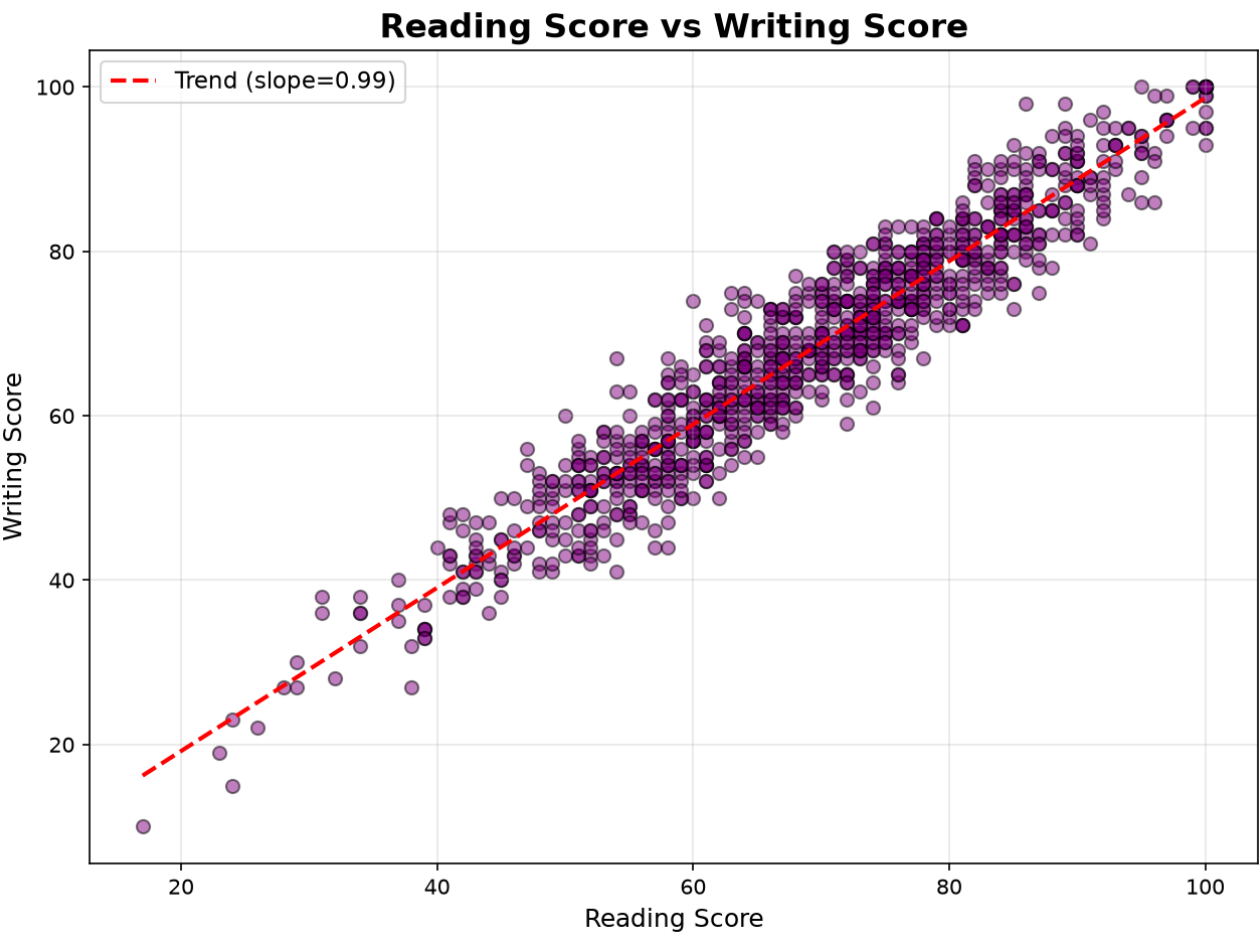
1. Histogram - Distribution of Math Scores



2. Bar Chart - Average Score by Subject

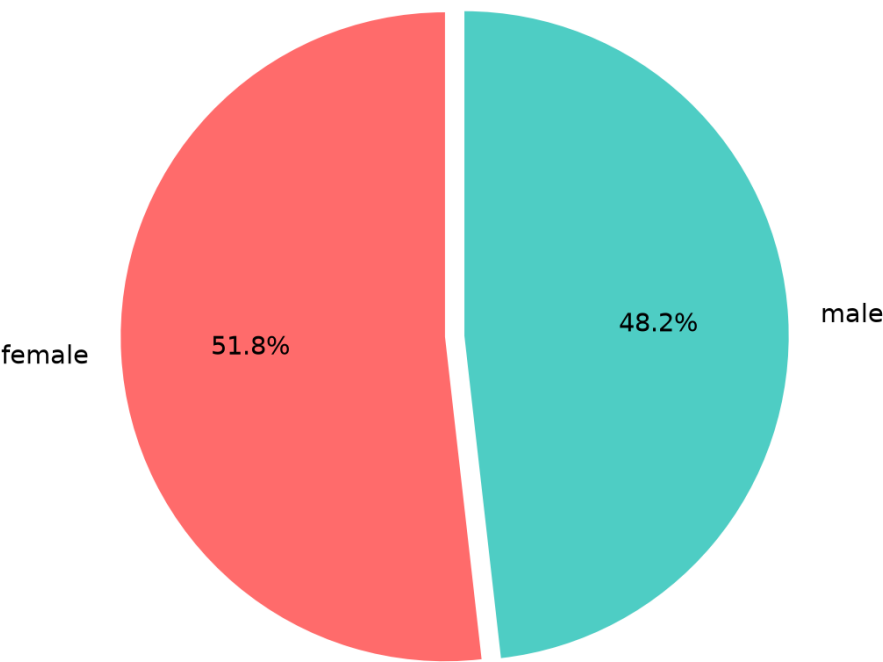


3. Scatter Plot - Reading Score vs Writing Score



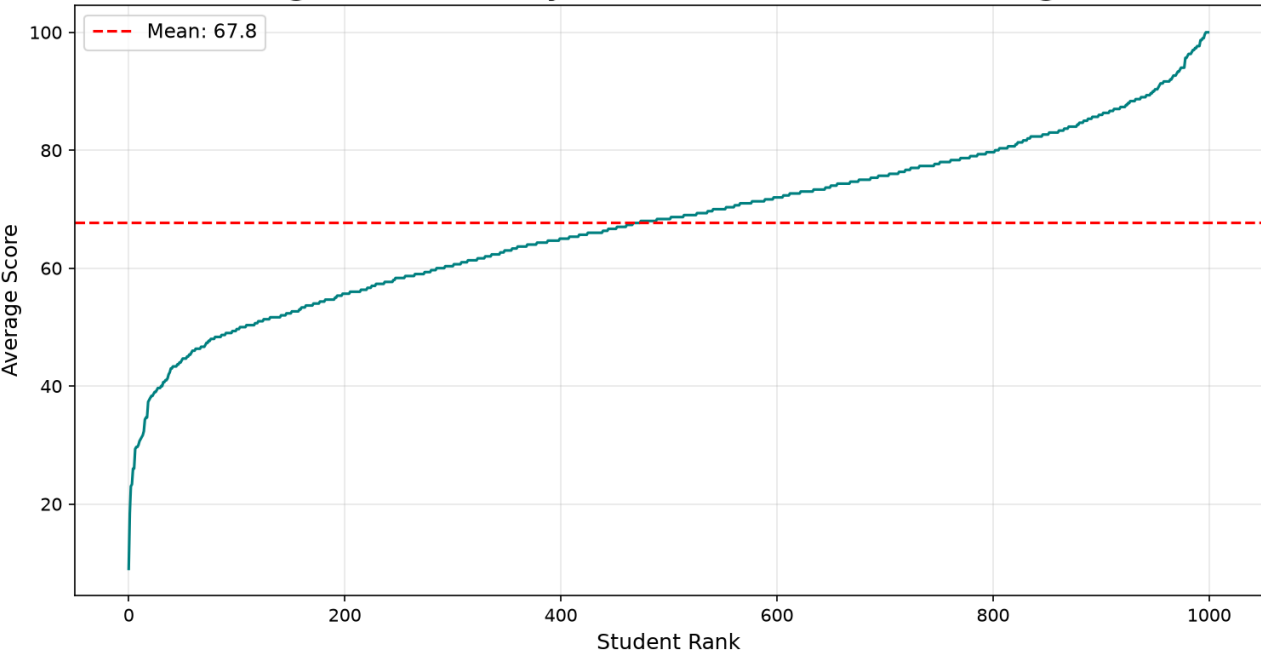
4. Pie Chart - Distribution of Students by Gender

Distribution of Students by Gender



5. Line Plot - Average Score of Every Student (Sorted)

Average Score of Every Student (Sorted Lowest to Highest)



Part F - Interpretation

1. Which subject appears easiest for students?

Reading has the highest average marks among all three subjects (approximately 69.17), followed closely by writing (approximately 68.05), and math has the lowest average (approximately 66.09). This suggests that reading appears to be the easiest subject for students overall, while math tends to be the most challenging. The higher standard deviation in math scores also indicates greater variability in student performance.

2. Does test preparation improve performance?

Yes, students who completed the test preparation course scored consistently higher across all three subjects compared to those who did not. On average, students who completed test prep scored approximately 7.63 points higher overall (72.67 vs 65.04). This indicates that test preparation has a significant positive impact on student performance, with improvements seen across math, reading, and writing scores.

3. Which visualization provided the most useful insight?

The scatter plot of Reading Score vs Writing Score provided the most useful insight because it clearly shows a strong positive correlation between the two subjects. The tight clustering around the trend line suggests that students who perform well in reading also tend to perform well in writing, indicating these skills are closely related. This visualization makes it easy to identify patterns and outliers in the data.

4. Suggest one improvement schools could make based on the data

Based on the data, schools should implement mandatory test preparation programs for all students. The data clearly shows that students who completed the test preparation course performed significantly better than those who did not. Additionally, since math scores are consistently the lowest across all demographics (mean 66.09 vs 69.17 for reading), schools should allocate more resources to math tutoring and support programs, particularly for students from lower parental education backgrounds who tend to score lower overall.